

Semialgebraic stratified stability for finite measurements

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Abstract

Alberti, Arroyo and Santacesaria ask whether their low-dimensional inverse problem theory can be extended to generalized settings with boundary points and variable parameter number, such as degenerate polygons. A fixed-dimensional boundary/corner extension is already recorded in a companion packet, while a frontier counterexample shows that arbitrary smooth stratified contact is too broad. This note gives a positive tame substitute: compact semialgebraic stratified parameter sets enjoy global Hölder stability from injective semialgebraic finite measurements. The proof is an immediate but useful application of the semialgebraic Łojasiewicz inequality on $P \times P$.

1 Source question

The source paper proves Hölder and Lipschitz stability for nonlinear inverse problems $F(x) = y$ when the unknown x lies in a compact subset of a finite-dimensional differentiable manifold M . In its conclusion it asks for extensions beyond manifolds without boundary and fixed dimension.

learning may be used to learn some of the parameters of the network, like those related to the manifold M which, as mentioned before, may be unknown.

- The manifolds considered in this work are without boundary and, by definition, of fixed dimension. However, there are situations where boundary points may be of interest, in order to cover, for instance, the case of degenerate polygons, for which the number of parameters vary. Extending the current theory to this generalised setting would enlarge the range of applicability of these results.
- Another key assumption of the main results is the injectivity of the differential of F . However, in some cases, the differential of F may not be injective. It would be interesting to investigate whether a workaround may be found by using higher order derivatives of F , as in certain higher order

The previous boundary/corner packet handles fixed-dimensional endpoint and corner charts. The harder issue is genuine stratification: a positive dimensional family can collapse to lower-dimensional strata.

2 Setup

Let $P \subset \mathbb{R}^m$ be a compact semialgebraic set. Think of P as a finite-dimensional parameter space that may have strata of different dimensions. Let X be a Banach space and let $\iota : P \rightarrow X$ be the parameter-to-unknown map. We assume that ι is α -Hölder for some $0 < \alpha \leq 1$:

$$\|\iota(p) - \iota(q)\|_X \leq L|p - q|^\alpha, \quad p, q \in P.$$

Let $G : P \rightarrow \mathbb{R}^N$ be a finite-measurement map. In applications one may have

$$G(p) = QF(\iota(p)),$$

where F is the forward map and Q records finitely many measurements. The theorem below assumes directly that G is continuous, semialgebraic, and injective on P .

3 Idea of the proof

The counterexample $e^{-1/t}$ fails because measurement separation from a frontier stratum is flatter than every power of parameter separation. Semialgebraic geometry rules out precisely this phenomenon. On a compact semialgebraic set, if a nonnegative semialgebraic function vanishes wherever another one vanishes, then the first is bounded by a power of the second. Apply this to the pairwise parameter distance and pairwise measurement distance on $P \times P$.

4 The theorem

Theorem 1. *Let $P \subset \mathbb{R}^m$ be compact and semialgebraic, let X be a Banach space, and let $\iota : P \rightarrow X$ satisfy*

$$\|\iota(p) - \iota(q)\|_X \leq L|p - q|^\alpha, \quad p, q \in P,$$

for some $L > 0$ and $0 < \alpha \leq 1$. Let $G : P \rightarrow \mathbb{R}^N$ be continuous, semialgebraic, and injective. Then there are constants $C > 0$ and $\beta > 0$ such that

$$\|\iota(p) - \iota(q)\|_X \leq C|G(p) - G(q)|^\beta, \quad p, q \in P.$$

Proof. Set $S = P \times P$. This is a compact semialgebraic subset of $\mathbb{R}^{2m}$. Define

$$f(p, q) = |p - q|^2, \quad g(p, q) = |G(p) - G(q)|^2, \quad (p, q) \in S.$$

Both functions are continuous, nonnegative, and semialgebraic. Since G is injective, $g(p, q) = 0$ implies $p = q$, hence $f(p, q) = 0$. Thus

$$g^{-1}(0) \subset f^{-1}(0).$$

By the semialgebraic Lojasiewicz inequality on compact sets, there are constants $A > 0$ and $\theta > 0$ such that

$$f(p, q) \leq Ag(p, q)^\theta, \quad (p, q) \in S.$$

Taking square roots gives

$$|p - q| \leq A^{1/2}|G(p) - G(q)|^\theta.$$

Therefore

$$\|\iota(p) - \iota(q)\|_X \leq L|p - q|^\alpha \leq LA^{\alpha/2}|G(p) - G(q)|^{\alpha\theta}.$$

The result follows with $C = LA^{\alpha/2}$ and $\beta = \alpha\theta$. □

5 Consequences

Remark 1 (Derivative injectivity is replaced by tameness plus injectivity). *The theorem does not assume that G has injective differential on every stratum. In the semialgebraic finite-measurement setting, global injectivity is already enough to force a Hölder inverse on compact parameter sets. This is especially useful near singular strata, where the derivative may lose rank even though the finite measurements still separate all admissible parameters.*

Remark 2 (Why the frontier counterexample is excluded). *The companion counterexample uses $F(t) = e^{-1/t}$ near $t = 0$. This is C^∞ on the open stratum and continuous at the frontier, but its frontier separation is flatter than any power. Such behaviour is not semialgebraic. The present theorem explains why the counterexample is a genuine smooth-stratified obstruction but not an obstruction in tame algebraic parameter families.*

Remark 3 (Relation to degenerate polygons). *If a degenerate polygonal family is encoded by a compact semialgebraic parameter set P , if the map from parameters to unknowns is Hölder into the ambient Banach space, and if a finite measurement vector G is semialgebraic and injective on P , then the theorem gives the desired Hölder stability. The hypotheses are intentionally stated at the finite-measurement level; the paper’s arbitrary Banach-valued forward maps need not be semialgebraic in any finite-dimensional sense.*

6 Scope and limitations

This is a partial positive result for the source’s generalized-setting question. It gives a full Hölder stability theorem for compact semialgebraic stratified parameter sets with injective semialgebraic finite measurements. It does not settle:

- arbitrary C^1 or C^∞ stratified frontier degenerations;
- Banach-valued forward maps without a finite-dimensional tame measurement representation;
- o-minimal structures with exponential functions, where flat behaviour like $e^{-1/t}$ may be definable;
- the source paper’s separate higher-order derivative and nonlinear compressed-sensing questions.

7 Novelty check

The cheap run indexes were searched for arXiv:2009.00574, “semialgebraic”, “stratified”, “Łojasiewicz”, and “degenerate polygons”. Existing hits for this source are the fixed-dimensional boundary/corner packet and the scoped frontier counterexample.

Web searches on 2026-06-28 for semialgebraic injective maps with Hölder inverse and Łojasiewicz inequalities found standard semialgebraic metric results, including Kurdyka–Spodzieja–Szlachcinska, but no separate later paper applying this directly to the stratified finite-measurement question from arXiv:2009.00574.

8 References

- G. S. Alberti, A. Arroyo, and M. Santacesaria, *Inverse problems on low-dimensional manifolds*, arXiv:2009.00574.
- K. Kurdyka, S. Spodzieja, and A. Szlachcinska, *Metric properties of semialgebraic mappings*, arXiv:1412.5088.